

ard_K210 Handwritten Digit Recognition

ard_K210 Handwritten Digit Recognition

1.K210 and Arduino communication

1.1 Experimental premises

1.2 Experimental wiring

1.3 Main code explanation

1.4 1.4 experimental phenomena

1.K210 and Arduino communication

1.1 Experimental premises

This tutorial uses arduino, and K210 requires running the program in **K210-AI (stm32_pico_arduino)** to start the experiment

arduino *1

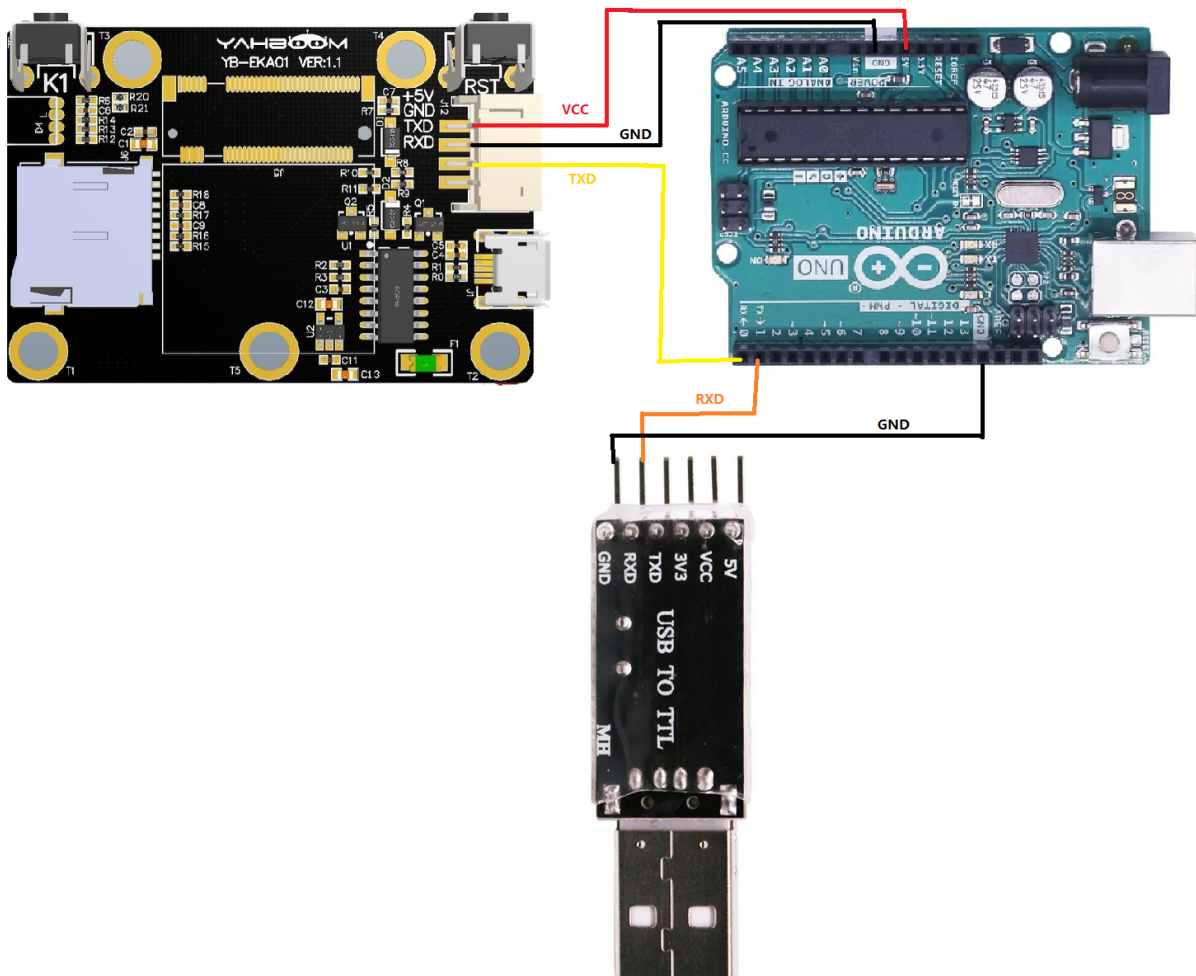
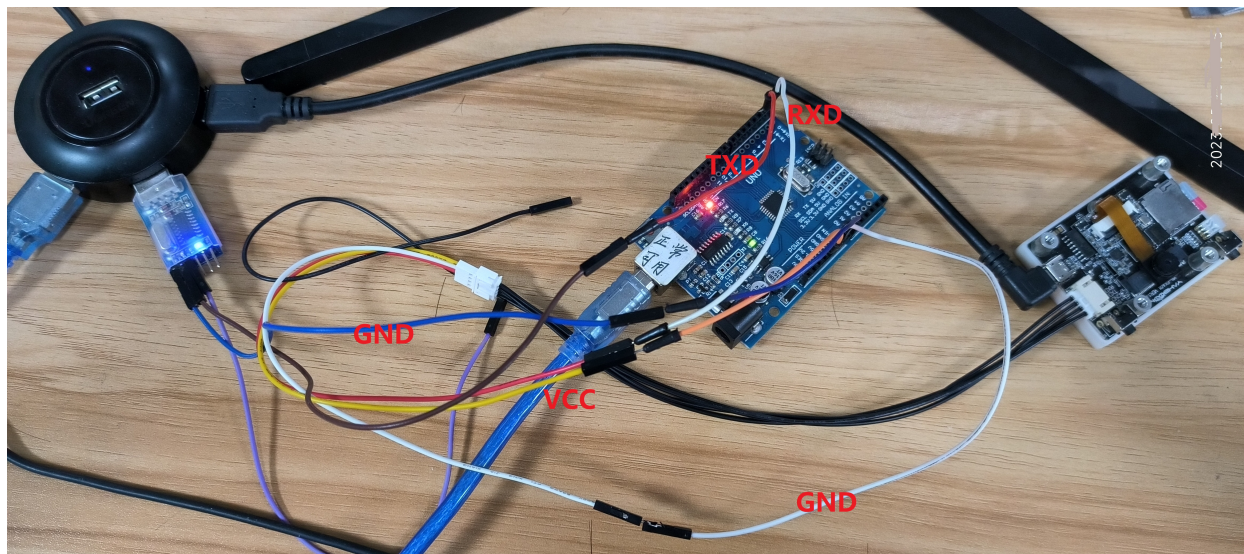
K210 perspective module * 1 (requires SD card (with AI model inside) and camera)

USB to TTL module * 1

1.2 Experimental wiring

arduino	usb to ttl
TXD	RXD
GND	GND

arduino	K210 perspective module
RXD	TXD
GND	GND
VCC	5V
Wiring as shown in the diagram:	



1.3 Main code explanation

```
void loop()
{
    while (K210Serial.available())
    {
```

```

recv_k210msg(k210Serial.read());

if (k210_msg.class_n != 0)
{
    if(k210_msg.class_n == 11)
    {
        sprintf(buff_com,"id = %c\r\n",k210_msg.id);
        k210Serial.print(buff_com);

        k210_msg.class_n = 0;
    }
}

}
}

```

After the above program, if you are running this routine, k210_ The members of the msg structure have corresponding values and are processed through serial port printing

K210_ Msg: is a structure that receives information, and its main members are

- X: is the horizontal coordinate of the top left corner of the recognized box (range: 0-240)
- Y: is the vertical coordinate of the upper left corner of the identified box (range: 0-320)
- W: is the width of the recognized box (range: 0-240)
- H: The length of the recognized box (range: 0-320)
- ID: is the recognized label
- Class_ n: Routine number
- Msg_ Msg [20]: Valid data

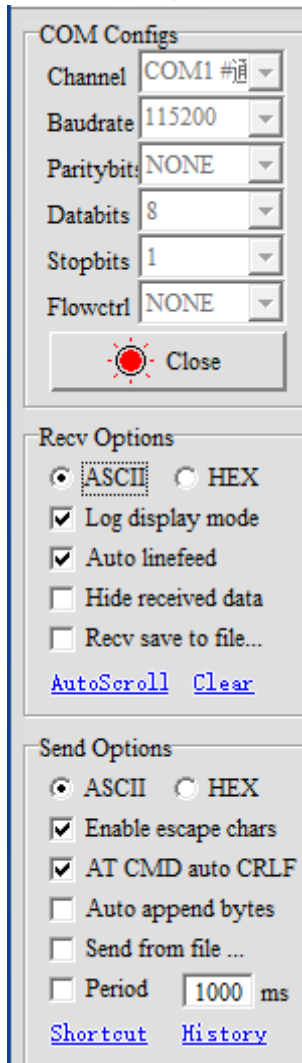
After receiving and processing data, k210_ Each member of the msg will store valid information.

If you want to develop it again, call K210 directly_ Members of msg are sufficient

1.4 1.4 experimental phenomena

1. After connecting the cable, the K210 perspective module runs offline. Please check 【6.2 K210 as coprocessor】 -- 【ReadMe】

2. Set the serial port assistant to the interface shown in the figure



The image shows a 'Serial Port Assistant' configuration window. It is divided into three main sections: 'COM Configs', 'Recv Options', and 'Send Options'. In 'COM Configs', 'Channel' is set to 'COM1', 'Baudrate' to '115200', 'Paritybits' to 'NONE', 'Databits' to '8', 'Stopbits' to '1', and 'Flowctrl' to 'NONE'. There is a 'Close' button with a red stop icon. The 'Recv Options' section has radio buttons for 'ASCII' (selected) and 'HEX', and checkboxes for 'Log display mode' (checked), 'Auto linefeed' (checked), 'Hide received data' (unchecked), and 'Recv save to file...' (unchecked). It also has 'AutoScroll' and 'Clear' links. The 'Send Options' section has radio buttons for 'ASCII' (selected) and 'HEX', and checkboxes for 'Enable escape chars' (checked), 'AT CMD auto CRLF' (checked), 'Auto append bytes' (unchecked), 'Send from file ...' (unchecked), and 'Period' (unchecked) with a value of '1000 ms'. It also has 'Shortcut' and 'History' links.

3. Then run the handwritten digit recognition routine, and the serial assistant will print out the important information transmitted from k210 to stm32, as shown in the following figure



The image shows a serial assistant output window with a black background and green text. The text displays a series of 'id' values being transmitted. The sequence starts with 'id = 0', followed by several 'id = 8's, then 'id = 0', 'id = 6', 'id = 5', 'id = 1', 'id = 9', 'id = 1', 'id = 5', 'id = 6', 'id = 0', 'id = 6', and finally 'id = 5'. A vertical scrollbar is visible on the right side of the window.

Handwritten digit recognition only transmits k210_ The id of the msg is a Member variable.
ID: Refers to the identified numerical ID (0-9)